
Discussion Paper

Intellectual Property and the Digital Economy: Five Issues for International Norms and Trade Rules

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Abstract: Over the past two centuries, the nature of innovation has evolved profoundly across many dimensions, including in the pace and volume of change, in the cost and institutional structure of innovation activity, in the enabling environment for innovation (which itself has evolved because of innovation), in the sources of demand for innovation, in the proliferation of entities engaged in innovation (including most recently the role of artificial intelligence as a patent-receiving inventor), in the distribution of its benefits, and in the nature of concerns that new technologies have raised. With the emergence of the data-driven economy, the pervasive, protean and permanent characteristics of digitized data, one of today's most important types of intellectual property, create fundamental new social and political conditions that will necessarily elicit social and political responses. And along the way, innovation and the system of protection for intangible assets that innovation produces became the central bone of contention in the trade and technology war between the United States and China (in which third countries are increasingly embroiled). Even as these profound changes unfolded, the basic instruments incentivizing innovation – the various forms of intellectual property (IP) protection – were essentially unchanged and indeed formally expanded, even as the amount of protected IP expanded enormously. Given that many key metrics of economic performance and social behaviour have changed in problematic directions during this period, and in light of the fact that in the innovation-intensive countries the economy has been transformed from one dominated by tangible assets to one dominated by (protected) intangible assets, the system bears reviewing. This note identifies five issues for the current international norms and trade rules insofar as they cover the intangibles economy – in particular IP and the digital economy.

Keywords: intellectual property, digital transformation, data, knowledge-based economy, data-driven economy, intangibles, technology, trade secrets

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1 Introduction and Background

As we prepare to “build back better” in the post-pandemic world, the nexus of intellectual property (IP), innovation and trade rules faces a set of vexing problems. Basically, innovation has changed the context in which innovation takes place, while the instruments put in place to incentivize innovation have remained essentially unchanged. There are numerous implications for the framing of rules, starting with the pace and volume of change, the changed cost structure of innovation, the changed enabling environment, and the changes in societal needs. Further, while not directly related to the framing of IP rules, there are important developments regarding the distribution of benefits of innovation and geopolitical and geoeconomic issues related to the technologies that innovation produces that affect the policy context in which IP rules are made.

As regards the pace and volume of change, innovation has gone through four reasonably distinct phases (Ciuriak and Ptashkina, 2021a).

In terms of pace, innovation has steadily stepped up the tempo. In pre-industrial times, innovation was sporadic and did not serve to meaningfully accelerate economic growth. With the industrial revolution, innovation became continuous because of several feedback effects: innovations enabled more innovations through recombination, driving a self-sustaining dynamic (see, e.g., Poincaré, 1908; Weitzman, 1998; Ciuriak and Curtis, 2015); and positive feedback loops emerged between basic science and applied research as solutions to practical problems established the basis for subsequent scientific theoretical developments, which in turn spurred further technical progress (Rosenberg, 1982 provides examples). The industrialization of research and development (R&D) enabled by information technology ushered in the knowledge-based economy, an event marked by the steep increase in the pace of creation of IP post-1980. The development of machine learning, which industrializes the very act of creation (which is symbolically underscored by the recent grants of a patent to an artificial intelligence entity by Australia and South Africa; Baker McKenzie, 2021; and Naidoo, 2021), turbocharged innovation in the modern digital and data-driven economy.

In terms of volume, the flow of innovation went from sparse to profuse over the course of these four phases. This reflects both the increase in human resources flowing to research and development within countries, the addition of major new innovation centres (in particular China and India in the last several decades), the steady cumulation of the knowledge base on which innovation builds, and the steady increase in the power of technology assisting innovation, culminating, of course, in machine learning (ML) and artificial intelligence (AI), which is massively scalable.

Along the way, the cost structure of innovation has changed, declining in some aspects and rising in others. In many areas, the cost of creation has been sharply reduced by enabling technology: viz. the shoestring 3D design labs and biofabrication labs that engaged in synthetic biology with far fewer resources than traditional labs (reported in Simons, 2012); and the enabling effect of access to high-end cloud computing services, including infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS), which combine to reduce entry costs for innovative companies even in advanced areas such as ML/AI applications (IBM, 2021). In many

other areas, costs have been virtually eliminated: viz. the creation of high quality images with ubiquitous cell phones, or TikTok videos, which allow any individual to become a creator and performer. At the same time, the cost of innovation has risen steeply in many areas of advanced science: Bloom et al. (2020) document a steep decline in research productivity – for example, the 18-fold increase in the number of researchers deployed today to sustain the doubling of computer chip density known as Moore's Law compared to the 1970s; they conclude that good ideas are increasingly hard to find (see also Miyagawa and Ishikawa, 2019, on the decline in research productivity). Knott (2017) looking at this same issue concludes that while established industries face increasing costs, innovative firms that create new industries do not.

The enabling environment for innovation in the information age has greatly improved, resulting in an evolution in the structure of innovation. In addition to traditional government and corporate labs, innovation today includes open-source collaborative technology development (Linksvayer, 2021); new forms of community-based innovation such as co-creation (Eckhardt et al., 2021); a continuum in the scale and significance of innovation starting with “grain-size”, sub-patentable, and follow-on innovations in areas where cumulative innovation matters (Ciuriak and Curtis, 2015); and a highly competitive environment in which next steps are largely baked in and the patent prize goes to the swiftest – or to the first researcher to be assigned the task (hence the race to patent). The introduction of AI as inventor is only the most recent of the transformative factors impacting on the nature of innovation.

Finally, the demand context for innovation has changed.

In wealthy countries, the main problems are hardly the lack of innovative private goods and services (there are issues of equitable distribution but these do not arise from scarcity of blue jeans or burgers) – they reflect major challenges in the public goods domain: the pandemic, climate change (and the weather-related disasters and agricultural dislocation that it is creating), the mass extinction underway in the biosphere that threatens the support ecosystem on which we depend (think bees), the greying of populations (think rising dependency ratios), and the plethora of social, economic and geopolitical governance challenges posed by the digital transformation. Innovation systems are front and centre in the response to these issues – and so therefore is the legal framework for innovation.

In developing countries, the main problem is not in creating new technology but in acquiring and adapting existing technology to raise living standards and address societal needs that also are mainly in the public domain – as a walk in the streets of any bustling city in the Global South will show, the immediate impression is not a lack of basic private goods but the poor quality of public goods (transportation infrastructure, quality of telecommunications, energy supply reliability, and efficient public administration). Acquiring new capabilities is inextricably entwined with acquiring companies that use technology. For developing countries, the problem of low technological capability is thus the flip side of the coin of missing companies. Fixing this problem will require developing countries to navigate the thicket of IP rules that create freedom to operate (FTO) for established firms (which are mostly domiciled in the advanced countries) – and deny such freedom to new firms.

Through the various phase changes, the structural evolution of the innovation system, and shifting demand environment, the basic instruments incentivizing innovation – the various forms of protected intellectual property (IP) – were essentially unchanged and indeed formal protection was expanded as trade secrets protection was upgraded to address the increased emphasis on this instrument for IP such as data and algorithms (Ciuriak and Ptashkina, 2021b). Moreover, IP reforms mostly worked to strengthen and lengthen protection, doubling down as it were on the model.

Given rising concerns about unintended consequences, it is not clear that doubling down on the model is what is required. Indeed, critics have voiced concerns about the emergence of what has been termed a new form of capitalism – “intellectual monopoly capitalism” (Pagano, 2014; Rikap, 2021). Moreover, technology and the IP and data that go with it have become the most important bone of contention in today’s geopolitics.

This note looks at five major issues facing the intangibles economy, including: the possible need for recalibration of the system of incentives, given the evolution of the nature of innovation; the need for consideration of systemic issues that impinge on the diffusion of technology to developing countries; the problems posed by intangibles for income redistribution through taxation; the unsettled state of governance of data in light of the myriad issues raised by datafication and the implications for international trade in digital and digitally-enabled modes; and the role of IP as a *casus bellum* in today’s geopolitical cold war.

2 Five Challenges for IP Norms and Rules

2.1 Recalibrating the Incentives for Innovation: Implications of the Changed Context

While necessity is the mother of invention, it would be hard to gainsay that the framework developed to incentivize innovation did not make an important contribution: we have witnessed a phenomenal technological transformation of human society over the past two centuries as a result of cumulative innovation, much of which, especially in the age of intangibles, depended on IP protection for monetization.

However, in recent decades, the bottom lines of the impact of all this innovation for societies and economic have been increasingly mixed. Key metrics of the impact on economies and societies have changed markedly:

- Product life cycles have shortened dramatically. While this has been a longstanding secular trend, the past decade has witnessed a palpable acceleration in the rate of product turnover. Horn (2020) observes that corporate revenues are now heavily dependent on new products, replacing a product or service line every two years is becoming the norm across many industries, and speed to market is critical if products are not to be already superseded by competitors.
- Intangible assets increased their share of returns to capital even as capital increased its share of national incomes, with profound consequences for income distribution and the

political economy of the innovating nations (see e.g., Haskel and Westlake, 2017; chapter six: “Intangibles and the Rise of Inequality”).

- Market concentration increased, a trend associated with the rise of “superstar firms” (Autor et al. 2020), which in turn can be linked to the role of intangible capital assets (Ayyagari et al. 2018).
- Evidence has mounted of waning business dynamism (see, e.g., Hathaway and Litan 2014; Economic Innovation Group, 2017; Rafieyan and Garcia, 2020). While this has been described as a case of “death by a thousand cuts” (Haltiwanger, 2015), features of the innovation system can plausibly be assigned part of the responsibility, including:
 - decreased freedom to operate for firms from IP proliferation reflecting strategic behaviour such as filing patent applications to pre-empt rivals (Gilbert and Newbery, 1982)
 - new business strategies of amassing patents for “assertion” in patent infringement suits rather than to support own production (FTC, 2016) – in the “Smartphone Wars”, companies were described as using patents as munitions (Lee-Makiyama and Messerlin, 2014);
 - reduced labour mobility in part due to non-compete clauses used to protect trade secrets (Lemley and Lobel, 2021); and
 - the interaction between market concentration, innovation and knowledge diffusion within an economy (Akcigit and Ates, 2019).

Accordingly, a case can be made that protection of IP worked all too well and started to generate negative impacts because of a failure to recalibrate and reform the instruments of IP protection in the face of the profound evolution of the system. In a nutshell, protection for intangible assets was developed in a world dominated by tangible assets and products. We now are in a world dominated by intangible assets. There appears to be a real question of balance.

2.2 Technology Diffusion and Economic Development

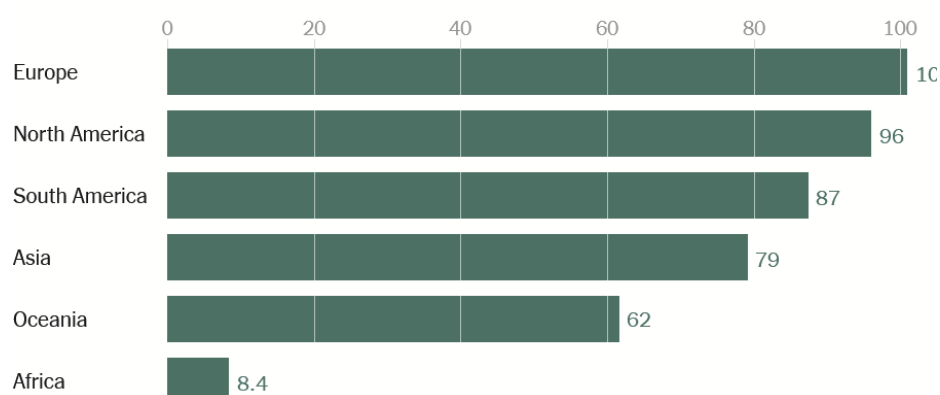
Internationally, the spread of IP protection through trade agreements – most importantly the World Trade Organization’s Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPs) – created *necessary* conditions for innovation in developing countries (a basic level of protection for IP), but not necessarily *sufficient* conditions. High quality trade agreements ratcheted up the WTO-mandated level of protection for IP, including through commitments to sign onto various conventions of the World Intellectual Property Organization (WIPO), introduction of more stringent enforcement (including criminal liability), and most recently commitments on cross-border data flow and restrictions on data localization requirements.

There is little in the way of evidence to support the argument that the spread of IP protection has helped innovation in developing countries. As noted, innovation involves technology and technology involves firms. Successful innovation leads to the emergence of high growth firms. One useful metric for this is the number of unicorns a country has. By one recent count of unicorns, only 39 of 811 were located in Africa, Latin America and Southeast Asia combined (CB Insights, 2021). The Global South is simply not unicorn habitat.

International norms and trade rules were framed against an understanding of trade and innovation rooted in classical theory and empirics in which countries trade based on comparative advantage, in which exchange is win-win, and in which government intervention distorts an otherwise level playing field. These rules and norms were not conceived against the understanding of trade and innovation in modern heterogeneous firms theory and empirics, in which it is understood that it is firms that trade – not countries – and that firms have widely varying levels of productivity and other capabilities, including in innovation (Ciuriak, et al., 2014). There is no “level playing field” conceptually in modern trade theory (Ciuriak, 2021a). And trade in the knowledge-based and data-driven economy depends on protection which enables monetization of intangible assets, and the role of IP protection in creating FTO and denying FTO to competitors. To lag in the modern economy is to lose, as has been profoundly made clear in the extraordinary inequality of developing countries – particularly those in Africa – in access to vaccines during the pandemic.

Vaccination rates by continent

Doses administered per 100 people



Source: Holder (2021).

It seems fair to conclude that the modern economy is not conducive to rapid technological convergence – and it is striking that the country that has stood out by its success in this regard – this is of course China – has come under extraordinary fire for its approach to acquiring technology. As Blumenthal and Zhang (2021) observe:

“... the U.S. Trade Representative estimated in 2018 that Chinese theft of American IP costs U.S. firms between \$225 billion and \$600 billion every year. General Keith Alexander, a former National Security Agency director, has called China’s technology theft “the greatest transfer of wealth in human history.”

While the analysis behind these claims has been challenged (see discussion below), the weak innovation dynamic in the Global South when considered in light of the pushback against China points to a systemic problem.

2.3 Remedial Re-distribution of Benefits: Intangibles and Taxation

Economic returns in the digital economy increasingly accrue to intangible assets like data, which are primarily captured by superstar firms headquartered in the leading innovation economies. The

reasons are fundamental and varied (Ciuriak, 2018a). The political economy of this system requires redistribution, which implies taxation.

However, intangible capital is particularly difficult to tax (Ciuriak and Eurallyah, 2021). Moreover, quite apart from issues such as use of tax havens to shelter income from intangible assets, there are conceptual issues: neither the slew of digital services taxes (DSTs) that have been proposed or implemented, nor the OECD/G20 Inclusive Framework, incorporate the value of data in their tax base or national allocation formulae.

By the same token, the value of data is not included in the estimates of international tax transfers that would underpin the determination of “fairness” of the final result of any agreed multilateral framework. Accordingly, the ability of developing countries to capture a share of data rents seems to be very limited (Ndajiwo, 2020).

This issue does not bear directly on the design of IP norms and rules but it does bear on our understanding of the impact of IP norms and rules on economic welfare.

2.4 Data Governance is Evolving

The pervasive, protean and yet permanent characteristics of digitized data create fundamentally new social and political conditions that will necessarily elicit social and political responses (Ciuriak and Wylie 2018; Ciuriak, 2019). Data measures in trade agreements will likely be compromised by these measures as countries respond to “data sovereignty” concerns (Aaronson, 2021), as well as cybersecurity and national security concerns (Ciuriak, 2021b). By the same token, the ultimate value and impact of data measures in trade agreements cannot be realistically evaluated at this stage – even though they may be the most important part of the trade agreements (Ciuriak, 2017a).

This is an important background issue for intellectual property norms and rules: in its study of the new US-Mexico-Canada Agreement (USMCA), the United States International Trade Commission (USITC) evaluated the measures on cross-border data flow and the ban on data localization as sufficiently large to flip the value of the agreement to the United States from -0.12% of GDP to +0.35% of GDP (USITC, 2019; Ciuriak, 2020a).

This issue does not bear directly on the design of IP norms and rules but it does bear on our understanding of the impact of IP norms and rules on economic welfare and on how IP will be handled in trade agreements in the future. When it comes to data regulation, authorities around the world are effectively “crossing the river by feeling for the stones,” moving forward in the absence of settled analysis and course-correcting as they go. This is particularly important for the evolution of trade secrets law as it covers data and related areas such as algorithms etc. Accordingly, where regulation will end up is far from clear.

Notably, in both in the USMCA and in the Digital Economy Partnership Agreement (DEPA) struck by Chile, New Zealand and Singapore, the GATT Article XXI national security exemption language is retained, but the examples are dropped, and no new examples are introduced. The fact that the drafters of these agreements have essentially punted on this issue leaves completely open

what kinds of risks to national security would support an abrogation of trade commitments in the digital age.

This underscores that data is not yet “treaty ready” (Ciuriak, 2018b) – and may not be for a while as the major economies scramble to address the plethora of issues the digital transformation is creating.

2.5 Intellectual Property and Geoeconomics

In the 21st Century, intangible assets protected by IP norms and rules have become the most important bone of contention between the great powers of the day, taking over this role from land in the pre-industrial era and mercantile dominance in the form of control of the sea lanes, ports and resources (especially oil) in the industrial era (Ciuriak, 2020b). The resulting geopolitical and geoeconomic rivalry has become a new “Great Game” as countries draw on their full repertoire of tactics (so far short of kinetic war) to gain the advantage (Ciuriak and Ptashkina, 2021c).

Recent scholarship underscores that concerns about leakage of IP through forced technology transfer, economic espionage, and trade secret theft, which are the core basis for the current trade and technology war between the United States and China (and in which third countries are increasingly embroiled), are very much inflated.

Sykes (2021) points out that the corporate structure requirements for investment in China, which are the main source of allegations of “forced technology transfer” are not violations of China’s WTO or other trade agreements; any technology transferred to Chinese joint venture partners as a result of these deals is consensual and, from the perspective of the Western firms, profit maximizing since they are not forced. Such corporate structure requirements may indeed be globally welfare enhancing.

Nowsrath (2021) examines economic espionage and observes that, in 2019, the US government identified seven spies who committed economic espionage or theft of trade secrets on behalf of China. Awards in US court cases involving trade secret theft are highly skewed, with the average settlement amounting to US\$ 5 million. While Nowsrath uses the highest award granted (\$250 million) to estimate the cost of these seven cases to the US economy, the more plausible estimate is based on the average, which places the cost at US\$ 35 million in 2019.

Ciuriak and Ptashkina (2021b) revisit the proxy method used to evaluate the scale of trade secret theft and conclude that the annual cost to the United States would be on the order of US\$ 2 billion. This estimate takes into account that most cases of trade secret would not actually be apprehended and litigated.

Both Nowsrath and Ciuriak and Ptashkina consider the gains made by the United States in its commercial relations with China. In the case of Nowsrath (2021), the metric is the spending in the United States by Chinese tourists, workers, and students, a figure which comes to US\$ 48.8 billion in 2018. In the case of Ciuriak and Ptashkina (2021b), the consideration is the US “brain gain” in innovation centres, which almost assuredly dominates any reverse flow of knowledge in the opposite direction (legal or otherwise), based on the significant share of the US research community with origins in China.

The weaponization of IP transfer is thus likely to be damaging to all countries – including those that are doing the weaponizing. While again this issue does not directly bear on the framing of specific rules and norms for IP, it is an important part of the background in which IP policy is made.

3 Discussion and Conclusions

The evolution of the economy in the last several decades from one based on scalable industrial production to one based on returns to intangible assets has been driven by an unprecedented wave of innovation that has, through complex feedbacks, changed the nature of innovation. The systems that were put in place in pre-industrial times, when innovation was sporadic and sparse, and that have been retained since with little fundamental change, can lay claim to some credit in fostering this explosion of creation. At the same time, a plethora of concerns have been raised about the nature of the modern economy that can be characterized as unintended consequences of success in incentivizing innovation. In particular, where the industrial economy was based on scalable production that tended to erode economic rents, the knowledge-based and data-driven economy depends heavily on protection of IP for monetization, which tends to create rents. Accordingly, the growing share of economic returns to intangible assets can arguably be related to developments such as increased market concentration, waning business dynamism in the advanced countries, increased skewing in income distribution, and disappointing outcomes in terms of technological convergence in developing countries. These issues are not going away any time soon given the likely continued increase in the share of income flowing to intangibles as AI-driven technologies come on stream, and the central role of technology and innovation in global geopolitics.

Whither the system of IP rules and norms?

Some degree of recalibration and reform seems unavoidable. The modern framework for protection for intangible assets was developed in a world dominated by tangible assets and products. We are now in a world dominated by intangible assets. In almost every possible regard, the nature of innovation has evolved beyond what could have been imagined at the time the original instruments of IP protection were introduced – product life cycles have shrivelled as innovation became continuous and profuse; new institutional frameworks (open source and co-creation) have emerged; and we are on the cusp of the shift of innovation into machine space, as marked in 2021 by the first grants of patents to AI systems.

Any reforms would have significant implications for the international diffusion of innovation capabilities (which has been disappointing as shown by the very low share of unicorns in the Global South), for the sharing of benefits of the digital economy given the difficulties of taxing intangible capital, for the governance of the digital economy, and for the geopolitical/geoeconomic contest for technological superiority currently being waged between the United States and China – a context that will be reshaped as China strengthens the protection of IP and gets into the strategic litigation game to capture rents (Sagami, 2021), as anticipated in Ciuriak (2017c)

The complex context in which IP rules and norms exist thus poses a very significant challenge to would-be reformers. But reform, it seems, is inevitable.

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